Higher National Diploma in Engineering (Electrical)
Forth Year, First Semester Examination – 2021
EE3215- Broadband Communication

Instructions to Candidates:	No. of questions: 06
Answer any Five (5) questions	No. of pages : 03
All question carry equal marks.	Time : Three (3) hours

1.

i. What is Broadband Communication? **(03 Marks)**

ii. Write four types of broadband connections. **(04 Marks)**

iii. What are the benefits of DSL? **(04 Marks)**

iv. What is Satellite internet? **(03 Marks)**

v. Write advantages & disadvantages of Satellite internet. **(06 Marks)**

[Total 20 Marks]

2.

i. What is meant by loading coil? Explain. **(04 Marks)**

ii. Which type of Internet access delivery does the speed of the network depend on the collective activity of the user sharing the same node connected to the internet service provider (ISP)? **(03 Marks)**

iii. What is the difference between DSL and ADSL? **(03 Marks)**

iv. xDSL uses digital encoding to provide more bandwidth over existing twisted-pair telephone lines. Many iterations of xDSL allow the phone to be used for data communication at the same time it's being used to transmit data.
Explain below xDSL varieties based on Up/Down maximum data rates and their applications.
a.) ADSL b) SDSL c) HDSL d) IDSL e) VDSL

(02x5 = 10 Marks)

[Total 20 Marks]

3.

i. Asymmetrical Digital Subscriber Line (ADSL) is a method to use the existing analog local loop lines for digital data transfer to and from the home. It shares the local loop bandwidth with the voice channel so that you can use the regular phone line while using ADSL. Draw the connectivity diagram of ADSL connection showing different blocks from the Central Office to Subscriber Premises. **(05 Marks)**

ii. Explain the function of each block in the diagram you drew for part (i) **(04 Marks)**

iii. What are the two types of modulation techniques used in ADSL Technologies? Give a brief description for each illustrating the spectrum of each modulated signal. **(06 Marks)**

iv. What are the transmission constraints on DSL due to Copper wire and how they affect the DSL signal transmission? **(05 Marks)**

[Total 20 Marks]

4.

i. Explain how does ADSL works after installations at customer premises **(04 Marks)**

ii. Draw the diagram of how to install the ADSL connection at customer premises (From wall socket to telephone and computer). **(07 Marks)**

iii. Write description about DSLAM **(03 Marks)**

iv. What are the reasons to get low speed broadband? **(04 Marks)**

v. Is ADSL better than LTE? Explain. **(02 Marks)**

[Total 20 Marks]

5.

i. What Does Broadband over Power Line (BPL) mean? Explain. **(05 Marks)**

ii. Discuses Asynchronous Transfer Mode Technology. **(05 Marks)**

iii. What are the advantage of ISDN? **(05 Marks)**

iv. What are the reasons people choose ISDN? **(05 Marks)**

[Total 20 Marks]

6.

i. Explain. Worldwide Interoperability for Microwave Access (WiMAX) is a wireless communication technology. **(05 Marks)**

ii. Discussed Applications of WiMAX technology. **(05 Marks)**

iii. What is HSPA? **(05 Marks)**

iv. LTE is a next generation mobile broadband technology. What are the benefits of LTE to the network operator and the end user? **(05 Marks)**

[Total 20 Marks]